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Constitutional Compliance: The Hidden Driver of Economic Growth?

Licencjat (bachelor) degree thesis
Field of the study: Finance, International Investment and Accounting

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Warsaw, May 2025

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Summary

The thesis presents an up to date issue related to the compliance with constitutions and its consequences for the economy. A various data is collected and used in the regression models, with the fixed effects estimator providing the most robust results. The findings show that adherence to constitutional law is significantly associated with higher GDP growth rates. This provides a new perspective on the topic and offers convincing evidence for policymakers to uphold constitutional principles as a means of promoting economic development.

Key words

constitutional compliance, GDP growth, political economy, economic growth
(zgodność z konstytucją, wzrost PKB, ekonomia polityczna, wzrost gospodarczy)

Field of the thesis (codes according to the Erasmus program)

Economics (14300)

Thematic classification

JEL codes: E02; K11; K38; K42; O43; P14; P48

The title of the thesis in Polish

Zgodność z Konstytucją:
Ukryty czynnik wzrostu gospodarczego?

Statement on the Use of Artificial Intelligence

During the preparation of this paper, the author used ChatGPT to improve the quality of the text in terms of style. The author of the paper verified the content and edited the text of the paper in accordance with its assumptions, presenting her own intellectual contribution and taking full responsibility for the final form of the paper.

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INTRODUCTION

Although the constitutionality of government actions has already been a subject of interest for political economy researchers (e.g. Voigt, 2021), relatively little attention has been paid to the consequences of constitutional compliance, that is respecting or disregarding constitutional law, for economic outcomes. The central question in my study is: What is the impact of constitutional compliance on economic growth, as measured by the growth of gross domestic product?

Despite an in-depth review of the literature, I did not find a study that directly addresses this relationship. Most of the previous research aims at examining the causes, not the effects of the disobedience, which makes developing the research in the opposite direction even more interesting. The results of such a study can provide valuable guidance for policymakers and fill an existing research gap.

By studying the work of scholars from different areas of political economy, I formulate the hypothesis that the constitutional compliance should have a positive impact on the economy, in particular economic growth. In the analysis, I used a variety of data and applied several estimation techniques, with fixed country effects being the most reliable one.

What is important, I study the impact of the compliance with "good" constitutional setting, meaning that in the de jure provision of rights only the objectively just and moral ones are included. To give an example, the freedom of speech can be viewed as the morally good right, while the permission for the slavery existence is certainly a morally bad provision and will be excluded from the analysis. To the best of my knowledge, such approach is usually used in this kind of studies (e.g. Law and Versteeg (2013) or Metelska-Szaniawska (2021)). This is likely because these rights naturally lead to better political and economic outcomes and no one would predict e.g. slavery to do so. This can be of course backed by, for instance, the work of Acemoglu et al. (2001), who show that the extractive institutions lead to the overall worse economic performance.

The structure of the paper is as follows. Section 1 presents an extensive literature review, covering topics from institutional economics to constitutional compliance. Section 2 provides an overview of empirical research on constitutional compliance and its measurement. Section 3 describes the data, emphasizing various indicators developed by scholars using distinct methodological frameworks. It also presents estimation results based on cross-section and panel data, assessing their alignment with my hypothesis. Finally, I summarize the results and provide supporting analyses in the Appendix.

CHAPTER I.

Literature Review

Before we dive into the econometric model I would like to guide the reader through the important steps to understand the relationship between adherence to constitutional principles and economic performance. Just as a historian examines the roots of a civilization in order to understand its present, we must explore what has been discovered to date about the institutions and principles of constitutional economics in order to fully understand their impact and the impact of constitutional compliance on today's economy. Firstly, I would like to discuss the institutions, what are they, and what evidence for their significance has already been provided. Then I want to present constitutions as an important part of the institutional structure. Ultimately, we will learn the condition necessary for them to be truly relevant for economic outcomes.

1.1. Institutional economics

As North (1990) emphasizes, constitutions are a fundamental type of institution because they are formally created rules that structure human interactions. If constitutions are institutions, then adherence to them is an essential aspect of the institution's functioning. To show why it is justified to expect the positive impact of constitutional compliance on economic growth, I would like to discuss the existing literature on the topic, starting with the following definition proposed by North (1990): "Institutions are the rules of the game in a society or, more formally, are the humanly devised constraints that shape human interaction.". We can divide them into formal (written) and informal ones (not written, embedded in conventions, values and norms accepted by the society), where constitutional text surely belongs to the former category. According to the above definition, the author points to the fact that institutions shape incentives of economic, social and political actors and secure their property rights. Furthermore, using the historical and everyday-life examples, such as the low transaction costs in the US housing sector partially ascribed to the prevailing policies, he argues that specific institutions can influence efficiency by altering the transaction and/or production costs. Comparing developed and developing countries, as well as given countries over time, in terms of how they cope with economic and institutional issues, he concludes that institutions play a significant role in relative development.

Similar view is shared by Acemoglu et al. (2008), who claim that institutions are one of the fundamental causes of economic growth differences. To confirm this hypothesis, authors refer to the evidence from the history of South and North Korea. They notice that these countries have the same origins, cultural background and are located in the same geographical area - right next to each other. For these reasons they can be treated as a natural experiment to show the impact of implementation of two different institutional systems. The result is that Republic of Korea has grown significantly under capitalist policies, while Democratic People's Republic of Korea has not grown much under communist institutions. Another example is the success of Botswana attributed by the authors to constraints posed on the politicians. In this publication Acemoglu et al. focused also on the possible change in the existing rules, which led them to interesting conclusion that to initiate growth it may be necessary to alter not only *de jure* political power. *De facto* power should also be changed, since old practices could result in a return to an inefficient starting point (and sometimes even this solution turns out to be ineffective due to the path-dependency). This shows how important the interplay between *de jure* and *de facto* policies is. In Acemoglu et al. (2019) same authors define inclusive political and economic institutions and state that such (as opposed to the extractive ones) ensure property rights protection, which goes in pair with the greater levels of investment.

The two above-mentioned articles captured yet another very well-known approach to institutions, namely the colonial approach. Acemoglu et al. (2001) believe that European settlers mortality rates can be used as an instrument to measure how different institutions contributed to the divergence of the GDP per capita levels in former colonies. They collect the data, *inter alia*, on output per worker, institutional differences, the fraction of the population of European descent in 1900, the mortality of European settlers and estimates for the mortality rates of bishops in Latin America from 1604 to 1876. Then, the authors conduct OLS regressions, instrumental variable regressions and use first- and two- stage least squares estimation finding a network of significant correlations connecting mortality rates of the settlers with the quality of current institutions. The latter, in turn, proves to be crucial for economic performance (measured by GDP per capita). Acemoglu et al. argue that places characterized by high mortality rates were often condemned to extractive institutions, while places with low mortality rates had a greater chance of introducing inclusive practices. This framework seems to explain why some colonies experienced growth and the others fell into poverty. However, one has to be careful not to interpret this analysis as establishing a causal relationship, as according to the authors, the correlation might work in both directions, since wealthier countries can afford better policies.

This perspective has been discussed by Rodrik (2008), who as well acknowledges the fact that economic development itself can influence what kind of policies are being implemented and that these effects can reinforce each other. Nevertheless, he agrees that strong institutions are key for the economic success as they provide a stable environment and encourage innovation. He draws attention to the fact that (at least at the time) many researchers used to oversimplify the reality. They did it by examining only one cause of institutional development, just as Acemoglu et al. (2001) considered only colonial history, or by ignoring behaviors such as corruption and simply assuming that the presence of rules in writing is equivalent or sufficient for their effectiveness in practice. To prove the necessity of departing from such simplifications author recalls the following observation. He states that the most important matter for investors is to feel that their property rights are protected. Surprisingly, empirical evidence from China and Russia showed that even though in 1990s law on property rights protection was superficially more appealing in Russia, investors considered China more reliable and trustworthy. Therefore, author again emphasizes the need to analyze not only the *de jure* aspect of the policy, but also the *de facto* realization.

The approach of Acemoglu et al. (2001) has also met with criticism by Glaeser et al. (2004). Using OLS regression estimation authors empirically prove mutual positive correlation between growth and institutional quality, raising concerns about causality. Moreover, the study uncovers two novel findings. Firstly, the results suggest that human capital brought by the settlers able to survive in the area was more likely to be the factor that allowed the colonies to grow relatively fast. Secondly, the de jure constitutional laws *per se* turn out not to have an impact on economic output. To explain why Acemoglu et al. reached contracting conclusions, the authors point out that the most commonly used measures of institutions focus on actual choices, i.e. their quality, rather than on the constraints posed by them (according to definition). That is to say, differences in research outcomes appear due to the divergence between written rules and the actual situation in the countries.

Conclusion from this part is the key for understanding the constitutional compliance impact on the economy. For now, we can see that what changes the economic outcomes are not the rights or constraints appearing in formal or informal form. It is rather the way in which they are executed in reality (or if they are executed at all). But why the results for institutional quality can be used as a prediction in the context of constitutional compliance? The answer is very simple – since constitution is a formal institution, constitutional compliance is a part of institutional quality. They are not the same, since the latter covers more than just constitutional

provisions. However, it may already provide us with the insight we seek. With this in mind, it is time to consider the constitutions themselves.

1.2. Constitutional economics

Taking the next step in our journey to understanding the importance of constitutional compliance, we move from institutions to a distinct stream of literature which focuses on constitutional economics. To further explore the topic, we must first recognize why are constitutions an important part of institutional structure.

Constitutions as “a set of rules which constrain the activities of persons and agents in the pursuits of their own ends and objectives" (Buchanan, 1977) define not only political but also economic limitations posed on the society. They shape the interactions between the economic agents who want to maximize their gains within a given setting. While the orthodox economics concentrates on finding the equilibria under these given limitations, constitutional economics focuses on the choice of constraints themselves. The precursor in this field, James M. Buchanan, describes constitutional economic analysis as the one that “attempts to explain the working properties of alternative sets of legal-institutional-constitutional rules that constrain the choices and activities of economic and political agents, the rules that define the framework within which the ordinary choices of economic and political agents are made.” (Buchanan, 1991).

The term ‘constitutional economics’ itself was introduced by Richard B. McKenzie, who first used it in order to describe a specific branch of the public choice study during the conference in 1982 (Buchanan, 1990) and is used interchangeably with the term ‘constitutional political economy’. Taking into account the origins of this concept it is no surprise that it refers to, among others, the theory of public choice. It is also closely linked with the thought of Adam Smith, Knut Wicksell, Friedrich A. von Hayek and the Ordo-liberal school of economics (Voigt, 1997).

Constitutional economics follows two main trends: normative and positive, which correspond to the questions: ‘how should it be?’ and ‘how is it?’, respectively. Of course, in the sense of constitutional political economy, these questions concern the rules and constraints, so in the normative light one could ask for example: what should the constitution include? or how should society choose the constitutional rules so as they are fair and efficient? The positive strand, by contrast, seeks to answer the questions such as: why constitutions differ among countries? (Metelska-Szaniawska, 2005).

Although my research focuses primarily on the positive approach, particularly the relationship between constitutional compliance and economic growth, it is useful to first present selected normative arguments, as they will help to frame the discussion and clarify the broader significance of constitutions before we turn to their empirical consequences.

A particularly insightful normative contribution comes from Elster (1994), who considers several functions of constitutions and the preconditions for them to be in force. Based on various historical and scientific sources he comes to the conclusion that constitutions matter for the economic performance for the following reasons. First of all, they hold incumbents accountable, especially when a mechanism to remove them from office is provided. It commits those in power to considering the most effective solutions that optimally allocate the resources and have a positive (or at least non-negative) impact on the sense of security among citizens. A further purpose they serve is assuring predictability, which encourages long-term planning, saving and investment. Protection against time inconsistency and short-term passions are another, somehow connected functions, as both emerge from the precommitment measures that spread out the government's resolutions over time, while bounding officials to obey and fulfill their promises. For these functions to be effective, however, certain conditions must be met. To guarantee accountability, constitutions should promote stability, no matter which social group represents the majority at the time. In other words, they should not be a subject of manipulation and ought to work as a tool to prevent the minorities from being omitted and exploited for the good of the elites. When it comes to the protection against short-term passions one should be cautious, since the excessive rigidities might have a negative effect on overall economy during the crises, wars or severe, sometimes unexpected, events (such as pandemics or climate disasters). These functions make constitutions influence economic efficiency and security, where some rights can increase one while reducing the other. Thus, they affect economic performance, and if they work as intended, the impact should be positive.

Having discussed the normative basis for why constitutions matter, we now turn to the positive approach, which seeks to empirically examine how constitutional rules influence real-world outcomes. For the purpose of my study I will present only selected works from this research area, however, for those seeking a comprehensive review I highly recommend the two extensive surveys by Voigt (1997, 2011).

A notable contribution to this field is the work of Persson and Tabellini (2004), who provide an overview of the empirical findings on the effects of two core constitutional elements, i.e. forms of government and electoral rules, on economic outcomes. Their earlier study (Persson and Tabellini, 2003), based on the data from 75 democracies in the 1990s confirms

that parliamentary systems are generally associated with growth-enhancing policies, while the negative effects of presidential systems are most pronounced in countries with weak democratic institutions, where safeguards against power abuse are limited or ineffective. Persson (2003) expands this research by incorporating the panel data along with adding non-democratic countries to considerations and shows that parliamentary democracy is in general superior to both, presidentialism and non-democracies in terms of encouraging the increase of productivity. What is more, based on the reviewed literature and quantitive analyses, Persson and Tabellini (2004) state that the reforms of constitutions imply the tradeoff between representation and accountability, where plurality rule is strengthening the latter contributing to lower political rents and decreased corruption but decreasing the representativeness of the political actors who wish to please the pivotal voters.

While the rules concerning the type of the political design in the country are difficult to get around and their impact on the economy is easier to analyze, many constitutional rules are more fragile. Juridical independence, property rights protection or freedom of speech are just the examples of rules that might appear (and often do) in constitutions, but are easily disrespected by the abusive politicians who see their gain in such disobedience with regard to the law. Still, it is important to understand the economic implications of these de jure provisions, even before turning to whether they are effectively enforced.

Lewkowicz and Metelska-Szaniawska (2022) pointed out to the previous research findings which showed positive impact of de facto rules enforcement on economic performance. E.g. Metelska-Szaniawska (2016) showed that de facto enforcement of laws putting commitment mechanisms and constraints on politicians into work contributes to the better enforcement of economic reforms in post-socialist counties (therefore more inclusive and growth-friendly economic environment). Given the de facto rules yielding such positive effects, the authors ask the question if de jure law has any impact on de facto realization. Before investigating this issue they also note that, for instance, Bjoernskov (2015) finds a negative direct relationship, holding other factors constant, between de jure property rights protection and economic growth in post-socialist countries. Similarly, other studies such as already mentioned Metelska-Szaniawska (2016) report either an insignificant or, under certain conditions, a negative direct association between de jure constitutional court independence and economic performance in post-socialist countries during the 1989–2012 period. The de jure features identified by Lewkowicz and Metelska-Szaniawska (2022) proved to be entirely irrelevant to the participatory aspects of democracy. However, in terms of competitiveness,

only one factor, namely the formal framework governing the competitiveness of participation, was found to have a significant impact, but with a declining trend.

These results, while being very surprising recall the already mentioned work of Glaeser et al. (2004), in which the authors show that de jure constitutional laws have no impact on economic growth of countries. Similarly, Feld and Voigt (2003) showed that while de facto judicial independence has a positive and significant impact on economic growth (robust to outliers and adding more control variables), de jure judicial independence lacks significance. These findings only confirm that de jure rules seem to not be sufficient to guarantee the expected, positive economic outcomes.

These findings raise important questions about the actual influence of constitutions on economic performance. While they are widely assumed to be foundational for institutional stability and economic development, the evidence on their direct effects, at least in terms of de jure provisions, remains inconclusive or even counterintuitive. But does this mean that constitutions are in fact not as important as the analysis of their functions for politics and economy suggests? The key is to examine actual outcomes and the gap between what is legally established and what happens in practice, which will be done in the next section of this thesis.

1.3. Constitutional compliance

I have deliberately omitted the finding with which I would like to start the this subsection of literature review - the conclusion emphasized by all, Lewkowicz and Metelska-Szaniawska (2022), Voigt (2003), Elster (1994) and Persson and Tabellini (2004), that constitutions do indeed matter for economic outcomes - provided that they are de facto respected. With this observation, we finally turn to the topic of constitutional compliance, which already appeared in previous considerations, although not named and/or left out by the assumptions. The so-called de jure–de facto gap refers precisely to this problem of non-compliance, i.e. the divergence between de jure laws (appearing in the text of the constitution) and de facto realization. Voigt (2021) defines it as the "non-congruence between provisions explicitly written down in the document called the constitution and the behavior of the top representatives of the various government branches such as cabinet members, legislators, and members of the country's top court(s)". So on the other hand, compliance is the degree to which actual political practice aligns with the constitutional text, serving as a critical indicator of constitutional effectiveness and legitimacy.

To understand the dynamics behind non-compliance, Voigt (2021) proposes a cost-benefit approach. Author explains that the more constraints are posed on executives (through constitutional text), the more tempting the vision of the disobedience is. If they are able to obtain greater benefits by behaving against the law than by acting in accordance with it, they will try to bypass the rules. Moreover, constitutional provisions that are misaligned with their surrounding context are unlikely to be effectively upheld. There are many reasons why the lack of respect for the constitution may be beneficial for the incumbents or why the implementation of formal law may be difficult or even impossible.

But why is constitutionality so important from the economic perspective? Why should we care about identifying the causes of non-compliance and creating conditions that promote adherence? Why does this issue deserve serious attention? Voigt (2021) claims that non-compliance negatively affects economy as well as society as a whole, e.g. by infringing human rights. Regarding the economic consequences, he states that such gap undermines the functions of constitutional text. *De jure* laws which are ignored by those in power fail to reduce uncertainty. This in turn impedes the incentives for investment and does not allow for development that otherwise could arise. Gutmann et al. (2021) present a complementary view from the opposite angle: the authors emphasize that compliance with the constitution promotes economic and political stability, thus encouraging investment. Taken together, these perspectives highlight that whether or not a government adheres to its constitution is predominant for economic growth. If citizens do not trust politicians and do not feel financially secure within their country, they are unlikely to invest or engage in long-term economic planning (Rodrik 2008). This can lower the rate of GDP growth limiting the country's ability to modernize and compete globally. This is exactly why upholding constitutional promises matters - it's a question of public trust, which is essential for a country's rapid and effective development.

Having established the theoretical importance of constitutionality – especially its role in shaping economic performance – it is crucial to examine how this issue has been addressed in the academic literature to date. The following section outlines this body of work, highlights its key conclusions, and indicates the direction needed to address the existing research gap on this important issue.

CHAPTER II

Constitutional compliance: overview of measures and empirical impact

2.1. Research to date

Scholarly attention has increasingly turned toward the problem of governments' disobedience with respect to the written law, although the emphasis so far has been on identifying its determinants. These include, among others: cultural factors (Gutmann et al., 2021), leader characteristics (Gutmann et al., 2024) and political polarization (Lewkowicz et al., 2024), as well as extreme events (Choutagunta et al., 2022) and the role of civil society (Lewkowicz and Lewczuk, 2023). Other contributing factors are the nation-state (Grajzl et al., 2024), geographic region, level of wealth of the country, civil war, ethnic fragmentalization and the degree of ideological character of constitution (Law and Versteeg, 2013). Institutional and historical dimensions also matter, such as experience with representative assemblies (Pavlik and Young, 2023), the age and comprehensiveness of the constitution, and the country's level of democracy (Metelska-Szaniawska, 2021). This wide range of determinants provides us a complex picture, yet the focus on causes has overshadowed an equally important question: what are the consequences of constitutional (non)compliance? Although research in this area is limited, it is of high importance and offers valuable insights. The following section presents the main findings.

The first attempt to investigate the impact of constitutional compliance on economic performance was conducted by Metelska-Szaniawska (2021) and concerned the post-socialist countries. The author showed that the size of the de jure - de facto gap directly affects the effectiveness of one of the core functions of the constitution, i.e. guaranteeing accountability. Constitutions provide credible constraints thanks to which politicians' behavior is more predictable. They reassure citizens that the promises made will be eventually fulfilled and that the law will be upheld. When incumbents fail to adhere to the constitution, these provisions no longer hold, and people may grow anxious of what is coming next. The sense of security is disrupted, promises become meaningless and fear regarding possible repeated violation of the law appears. What is more, the author proved empirically that de jure-de facto gap has a significant negative impact on performance in economic reforms, including privatization, liberalization, enterprise restructuring and competition policy. This means that if a country with extractive economic institutions aims to introduce the inclusive ones, the government has to

strive for strict obedience to the constitution. Otherwise, the efforts put into the reform process will be wasted.

The second and most recent attempt to examine the economic consequences of constitutional non-compliance, conducted by Lewczuk and Metelska-Szaniawska (2025), extends beyond the scope of post-socialist countries. The authors adopted a dynamic panel data model with the standard within estimator and dynamic panel GMM model with the Arellano-Bond estimator for 175 countries covering the period 1900-2000. The results showed that higher levels of GDP per capita are associated with greater constitutional compliance. However, when tested by the countries' level of democratization and level of wealth this positive effect is significant only in case of high-income autocracies and low-/lower-middle-income democracies. Lewczuk and Metelska-Szaniawska further demonstrate that the legal category in which compliance seems to be exceptionally important to enhance economic development is property right protection and the rule of law. This aligns with our earlier discussion, where we identified compliance in this particular area as sufficient to make investors feel secure, which translates into greater economic activity. Authors confirm this observation, showing that the biggest contribution to the higher GDP per capita level caused by compliance is through foreign direct investment. When focusing solely on the property right protection indicator, overall investment also becomes relevant.

These two studies (to the best of my knowledge) represent the only existing research addressing the economic consequences of disregarding constitutional law. Presented results clearly indicate that non-compliance undermines the health of the economy by reducing the effectiveness of reforms, depriving politicians of trust and weakening the functions of the constitutional text. Additionally countries affected by law violations tend to reveal weak economic performance in terms of GDP per capita in contrast to those which stick to the constitutional rules. This is therefore a topic that should undoubtedly be developed and deepened.

To conclude, large body of literature, starting from the one concerning foundations of institutional economics, through considerations about the importance of the constitutional text for the economy, finishing with the latest research on constitutional compliance, supports my prediction that de jure - de facto gap might have a detrimental effect on economic performance. Previous investigations have established correlation between compliance and outcomes in terms of GDP per capita. Yet, little is known about its impact on the GDP growth. This is the trajectory I intend to pursue in my research. I would like to focus in particular on the effects concerning the effects on GDP growth. However, before proceeding to the empirical estimation,

it is necessary to get acquainted with the existing databases and the methods of calculating constitutional compliance in order to choose the one that best suits the needs of this study.

2.2. Measurements of constitutional compliance

Over the past two decades researchers have started to construct the first databases aimed at measuring constitutional (non-)compliance. These databases enable the examination of its underlying drivers as well as broader implications.

The first-ever database to measure constitutional compliance was developed by Law and Versteeg (2013), who aimed at ranking countries based on their performance. Their novel dataset was structured as follows: the authors began by gathering de jure data, which required analyzing the content of 729 constitutions adopted by 188 different countries through the years 1946-2010. For de facto data they use reports issued by Amnesty International and the U.S. State Department, CIRI indicators (by Cingranelli and Richards), Minorities at Risk Project's measurements, Oona Hathaway's coding for fair trial rights and World Bank's Development Indicators. Based on these, they created 15 de jure and 15 de facto indices divided into 3 main categories: personal integrity rights, civil and political freedom, as well as socioeconomic and group rights. De jure index of a specific right equals to 1 if it appears in the constitution and 0 otherwise. De facto index takes 3 values: 0 if the right is violated, 1 if it is fully protected and 0.5 in case of partial violation. The de jure-de facto gap is then measured as a 'constitutional underperformance score' which equals to the sum of points granted for de facto performance of each right that scored 1 for de jure protection, divided by the total number of points that can be earned for these de jure rights.

Yet, the first attempt to investigate the impact of constitutional compliance on economic performance conducted by Metelska-Szaniawska (2021) was performed using a different measurement method. Metelska-Szaniawska developed her own database focusing on seven civil and political rights in post-socialist countries. She calculates the size of the gap by subtracting de facto from de jure indicators (all in the range from 0 to 1). Thus, overall indicator is between -1 and 1, where positive and negative values correspond to under- and overperformance, respectively. Unfortunately, despite the fact that this database is very interesting, it does not suit my kind of research, since it does not take into consideration whether rights protected *de facto* are also those protected the jure. That is to say, e.g. given the overall index of 0 in the Metelska-Szaniawska approach, we know the number of promises contained

in the constitutional text is equal to the number of rights that are actually enforced, but it does not mean that the promised rights are the same as those implemented.

Regarding the database used by Metelska-Szaniawska (2021) I think it is right to mention also the database constructed by Mataic and Finke (2019) in a similar way (by subtracting the measure of restrictions covering religions from the measure of promised freedoms). Accordingly, this database also does not match specific de jure and de facto rights. In addition, it only examines religious freedoms, which makes it even less attractive than, for example, the Law and Versteeg database taking into account as many as 3 broad categories of rights. For these reasons, and because the results of the study only concern determinants of noncompliance in this limited scope, I decided not to go into the details of this research. However, I felt it was my duty to mention the existence of this intriguing database, which may be useful for those interested in the government's approach to the matters related to religion.

Finally, the most recent database on this topic is The Comparative Constitutional Compliance Database (Gutmann et al., 2024). This is the data, which will be used in my own empirical study, presented below. De jure data used to construct it comes from the Comparative Constitutions Project (Elkins et al., 2009), while the corresponding de facto records originate from Varieties of Democracy project (V-Dem; version 12).

Gutmann et al. (2024) propose 14 constitutional rules that are covered by one or more of such indicators. Compliance is coded 1 if the rule is present in the text and applied in reality, 0 if the rule given in the constitution is violated and 0.5 if the rule is not included in the constitution (regardless of the fact weather it is implemented or not). Then the rights are divided into four categories: political rights, basic human rights, civil rights, and property rights and rule of law. The aggregate indicators for each of these categories are derived using factor analysis, with the first factor extracted as the desired indicator for that category. All of them are then standardized to have mean equal to zero and standard deviation equal to one. Of course, the higher their value the greater the compliance level. Since these concern the “good” laws, i.e. morally right ones (as discussed in the introductory part of this thesis), the higher score also shows greater de facto inclusiveness. Indicators, in their final form, appear in the database named as: cc_prop, cc_polit, cc_civil, and cc_basic. Under the name cc_total we can find the indicator for 14 rights altogether, while dj_index constitutes the count of de jure rights occurring in the constitution and ranges between 0 and 14.

For the comparison purposes, authors decided to provide the following additional data: the indicators created with usage of the same de jure and de facto data but decoded so as to match the Law and Versteeg (2013) approach (cc_prop_lv, cc_polit_lv, cc_civil_lv,

cc_basic_lv and cc_total_lv), as well as V-dem indicator obtained through expert's evaluations (v2exrescon). Gutmann et al. (2024) argue that V-dem measure is in fact ambiguous, since little is known about the criteria used. It might also be highly biased due to the experts' own interpretations and subjectivity. Therefore I will not include it in my study. The extended version of CCCD database incorporates de jure and de facto data before aggregation.

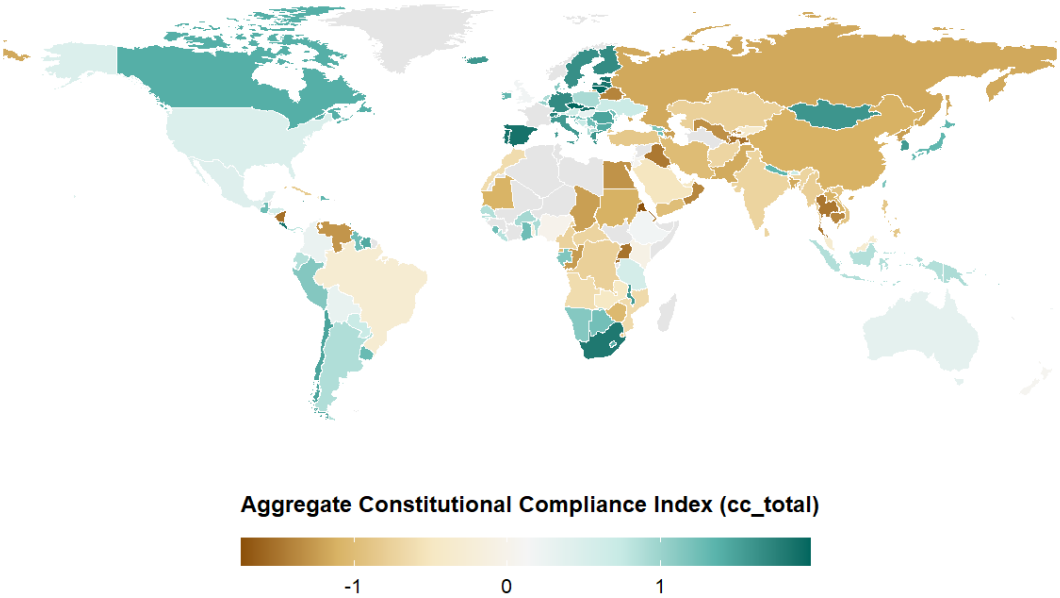
In the introductory article Gutmann et al. (2024) present several stylized facts about this database, some of which I would like to discuss briefly. First of all, what is clearly visible from the graphs provided by the authors, is that while aggregate constitutional compliance indicators (cc_total and cc_total_lv) behave more or less alike, the V-Dem indicator seems to follow the same trends only in short-term but diverge in the long-term. This observation confirms that v2exrescon yields different results than the other measures and, because of its questionable reliability, should be used with caution. Regarding the regional trends, it can be concluded that in 2020 African and Asian countries, in contrast to the European ones, experienced the lowest constitutional compliance levels. With the use of correlation coefficient calculated between GDP per capita and constitutional compliance (which took a low value of 0.26), authors reject the possibility that such result has to do with the income levels of these countries. Then, it is proved that democracies in general perform better than the other regime types. Specifically, incumbents in parliamentary and mixed democracies tend to respect rights written in constitution the most, while military dictatorships suffer the lowest compliance levels. Moreover, by grouping the countries according to their legal origin we see, that ones with German and Scandinavian civil law are doing much better than the others. On the other hand, when we look at the countries in terms of their colonial origin, it is evident that former colonies are doing worse than areas never colonized.

The above facts serve to highlight various tendencies for the specific groups of countries. When I will examine the impact of the existing de jure - de facto gaps on economic growth, I will also be interested in the correlations between GDP growth and compliance measures for some of these country groups. Therefore, the division categories proposed by the authors will be useful source of inspiration for my further research.

Following the stylized facts provided by the database authors, Pic 1 and Table 1 offer a practical illustration of how constitutional compliance looks in different parts of the world. Pic 1 presents a world map with the most recent data from 2020, showing how well countries follow their constitutions (cc_total index). Table 1 highlights the top three and bottom three countries in selected years, allowing for a comparison over time. The results show that stable democracies, particularly in Europe, tend to have the highest levels of constitutional

compliance. In contrast, countries affected by conflict or authoritarian rule, especially in Africa and Latin America, often score the lowest. These patterns reinforce the link between political regime type and the degree of constitutional adherence.

Pic 1. Data on the worldwide aggregate constitutional index values in 2020.



Source: Own study, based on Gutmann et al. (2024)

Table 1. Top and bottom 3 countries by cc_total in selected years - 1900, 1990 and 2020. Index Range: Maximum = 1.9723336, Minimum = -1.805778.

Top 3		**Bottom 3**	
Country	**cc_total**	**Country**	**cc_total**
1900			
Belgium	0.5693057	El Salvador	-1.279275
Denmark	0.5161524	Guatemala	-1.118224
Netherlands	0.4776889	Nicaragua	-1.049376
1990			
Portugal	1.8726763	Afghanistan	-1.805778
Spain	1.8726763	El Salvador	-1.645920
Costa Rica	1.8557690	Equatorial Guinea	-1.541468
2020			
Czech Republic	1.9723336	Eritrea	-1.733874
Lithuania	1.9723336	Nicaragua	-1.535040
Seychelles	1.8964918	Rwanda	-1.529356

Source: Own study, based on Gutmann et al. (2024)

CHAPTER III

Empirical modelling: explaining GDP growth with constitutional compliance

3.1. Data Overview

At this stage of the study, it is appropriate to present the specific variables included in the research. The study will involve both: cross-sectional as well as panel data regression aimed at estimating the parameters of an equation for economic development, measured as GDP growth. The goal is to examine the possible impact of the degree of adherence with constitutional text on the dependent variable. Below is the outline of the key variables that will be incorporated, starting from indicators of constitutional compliance. Then, other independent variables selected based on macroeconomic theory will be introduced.

3.1.1. Data on Constitutional Compliance

Compared to other available databases, The Comparative Constitutional Compliance Database (Gutmann et al., 2024), described in part 2.2 of this work, offers several advantages that make it the most appropriate choice for this study. It stands out from the other databases as, in addition to the fact that it takes only underperformance into account, it is based on de jure and de facto indicators all coming, respectively, from the same source (in contrast to Law and Versteeg database). Furthermore, its recency ensures that the analysis is based on the most up-to-date information available, which is crucial for drawing relevant and timely conclusions. For these reasons, I have chosen to rely on the CCCD database in my analysis.

It is important to note that the indices it provides capture constitutional compliance only with those rights that are present in a country's constitution (given the coding presented in section 2.2.). Thus, it reflects not only whether rights exist de jure, but crucially whether those promises are respected de facto. An increase in the index could reflect either: an increase in de facto respect for existing rights, or an increase in the number of rights included in the constitution and respected in practice. Importantly, simply adding rights to the constitution without ensuring they are followed can lead to a lower score than omitting them altogether, since the former is scored as 0, while the latter receives a score of 0.5. To conclude, a higher score indicates that a greater number of constitutional promises, whether long-standing or recently added, are being fulfilled.

In my study I will use all of the five indicators - four for each category of laws and the aggregate. Here is how the categories break down: cc_prop accounts for property rights, judicial independence, equality before the law and rule of law; cc_basic relates to right to life, freedom from slavery and protection from torture; cc_polit corresponds to freedom of association, freedom of assembly and the right to form parties; and cc_civil covers free media, free speech, free movement and religious freedom. All of these indicators, along with the overall measure of compliance vary from approximately from -2 to 2. Additionally, I will include the aggregate indicator developed using the method proposed by Law and Versteeg (2013).

What should also be said explicitly, is that I will follow the approach of Lewczuk and Metelska-Szaniawska (2025) and include the constitutional compliance indices in a logarithmic form in my regression models. Due to compliance measures falling into the range (-2,2) I have to transform them before applying logarithm by adding 2 and therefore getting rid of the negative values.

3.1.2. Other determinants of GDP growth

In addition to the indicators measuring the extent of compliance with constitution, I also incorporate other factors that influence economic development in the model. The selection of these explanatory variables was guided by Barro's (2001) research, which thoroughly examined the determinants of GDP growth. I would like to briefly explain the reasoning behind their inclusion in my study.

Firstly, let's discuss the initial level of GDP per capita. It enters the model in the logarithmic form and is accounted for due to the potential convergence, i.e. the faster growth of poor countries with respect to the rich ones. This conclusion might be driven from the Solow growth model, where the convergence appears as a result of the decreasing marginal returns to capital. However, since there is no reason to predict the absolute convergence we have to rely on the conditional one, where it occurs only if countries possess similar characteristics such as technology or institutions (Barro 2001). If we refer to the neoclassical models, e.g. already mentioned Solow-Swan model, the similarity between countries is demonstrated by a comparable level of parameter A in equation:

$$y = Ak^{\alpha}$$

where:

y - the GDP per effective worker

k - the level of capital per effective worker and

α - a parameter, $\alpha \in (0,1)$.

Thus, we have to include other variables that would capture the differences between the countries, so as the initial level of GDP per capita is significant for our results.

One of such variables might be the initial state of human capital in a given country. As human capital is inherently difficult to measure, we may use an instrumental variable as a proxy. In my model, I used an indicator that describes the level of educational attainment for this purpose, specifically, the school enrollment rate at the tertiary level (measured as the percentage of the population within the age group corresponding to this level of education). We might also include the fertility rate, which is expected to have a negative impact on human capital (because parents have less time to pursue their careers and engage in higher education due to the responsibilities connected with taking care of the children) and therefore it should also have a negative impact on economic growth. Another important factor - the health of the population - is reflected in the initial life expectancy at birth (measured in years).

Another variable of interest is the government expenditure. According to Barro (2001), government consumption, especially the unproductive spending that does not target sectors like education or defense, has a negative impact on economic growth. This is mainly due to its association with the distortionary taxes, reducing the disposable income of investors. Investment, in turn, plays a crucial role in the economic development as its higher values increase the stock of accumulated capital. To better isolate the growth-reducing component of government spending, I construct the variable gov_exp_reduced, which represents total government expenditure net of spending on education and defense.

To show the possible influence of the market characteristics on the GDP growth, I include the variable measuring the market distortion – inflation – and the measures of international openness – terms of trade and the ratio of exports plus imports to GDP, which enter the model individually and as an interaction term. Additionally, the V-Dem Index for liberal democracy is used to see if the type of regime plays a significant role for growth or the constitutional compliance influence on growth.

Table 2. presents the abbreviated names, full names and data sources of all variables used, while descriptive statistics are presented in Appendix 1 Table A1 (these include the number of observations, minimum and maximum value, mean, median and the standard deviation).

Table 2. List of the variables used in empirical model.

Variable	Description	Data source
Dependent variable		
GDP_growth	GDP growth (annual %)	World Bank (2025)
Constitutional compliance indicators		
cc_total	Aggregate indicator of government compliance with 14 constitutional rights	Gutmann et al. (2023)
cc_total_lv	Aggregate indicator of government compliance with 14 constitutional rights, calculated using the method proposed by Law and Versteeg (2013)	Gutmann et al. (2023)
cc_prop	Indicator of constitutional compliance: property rights, judicial independence, and rule of law.	Gutmann et al. (2023)
cc_basic	Indicator of constitutional compliance: right to life, freedom from slavery and protection from torture	Gutmann et al. (2023)
cc_civil	Indicator of constitutional compliance: free media, free speech, free movement and religious freedom	Gutmann et al. (2023)
cc_polit	Indicator of constitutional compliance: freedom of association, freedom of assembly and the right to form parties	Gutmann et al. (2023)
Other determinants of GDP growth		
fertility	Fertility rate, total (births per woman)	World Bank (2025)
GDPpc2015	Initial GDP per capita (constant 2015 USD)	World Bank (2025)
gov_exp	General government final consumption expenditure (% of GDP)	World Bank (2025)
gov_exp_edu	Government expenditure on education, total (% of GDP)	World Bank (2025)
gov_exp_milit	Military expenditure (% of GDP)	World Bank (2025)
gov_exp_reduced	Total government expenditure reduced by expenses on education and on defense (% of GDP)	World Bank (2025)
inf_def	Inflation, GDP deflator (annual %)	World Bank (2025)
investment	Gross capital formation (% of GDP)	World Bank (2025)
life_exp	Initial life expectancy at birth (years)	World Bank (2025)
education	Initial school enrollment, tertiary (gross %)	World Bank (2025)
tot_growth	Growth of terms of trade calculated from 'Net barter terms of trade index (2015 = 100)'	World Bank (2025)
trade	Trade (% of GDP) - the ratio of exports plus imports to GDP	World Bank (2025)
rule_of_law	Rule of Law: Estimate, standardized	World Bank (2025)
ctrl_of_corr	Control of Corruption: Estimate, standardized	World Bank (2025)
gov_eff	Government Effectiveness: Estimate, standardized	World Bank (2025)

Variable	Description	Data source
pol_stab	Political Stability and Absence of Violence/Terrorism: Estimate, standardized	World Bank (2025)
reg_q	Regulatory Quality: Estimate, standardized	World Bank (2025)
voice_and_acc	Voice and Accountability: Estimate, standardized	World Bank (2025)
v2x_libdem	The extent to which liberal democracy is realized in a country, reflecting how well its principles are upheld	V-Dem, Coppedge et al. (2024)

Source: Own study

3.2. Estimation results and hypotheses verification

I developed two models: the first is based on cross-sectional data from the period 2010-2019, while the second uses panel data spanning 1990-2020. I will present these models one at a time, beginning with a simple OLS regression. In the following section, I will then present the results of a fixed effects estimation using the Driscoll and Kraay (1998) robust standard errors.

3.2.1. Cross-sectional estimation

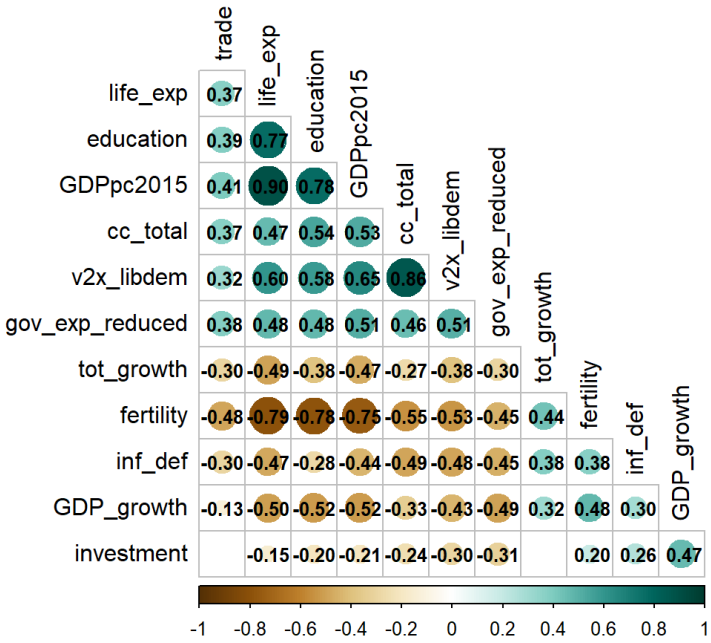
I began my cross-sectional analysis with model 0, which includes all of the explanatory variables discussed in the section 2. My goal was to check how the model performs under different conditions by adding or removing certain variables and to assess the variance inflation factor to determine if there is significant multicollinearity among them. At this point I would like to highlight that the significance level I adopt in my study is 5%, unless stated otherwise.

$$\text{GDP_growth} = \beta_1\times\text{fertility} + \beta_2\times\log(\text{GDPpc2015}) + \beta_3\times\text{inf_def} + \beta_4\times\text{investment} + \beta_5\times\text{life_exp} + \beta_6\times\text{education} + \beta_7\times\text{ tot_growth:trade} + \beta_8\times\text{gov_exp_reduced} + \beta_9\times\log(\text{cc_total}+2) + \beta_{10}\times\text{v2x_libdem} + \beta_{11}\times\text{ tot_growth} + \beta_{12}\times\text{trade} \quad [0]$$

It was proved useful to remove the life expectancy and education from the model. The first one had VIF equal to 9.23 indicating severe multicollinearity and being a source of a strong risk of redundant predictor affecting model stability, whereas the second variable was substantially decreasing the number of observations, due to limited sample size, which given the large number of variables (12) again was destabilizing the model and decreasing its accuracy. Moreover, I decided to also exclude the V-Dem democracy index as it appeared to

be highly correlated with our variable of interest – cc_total. Pic 2. presents the correlation coefficients between all of the variables from model 0, where the places with the calculated coefficient not being statistically significant are left blank. It is also worth mentioning that I used the Spearman's correlation calculation method, which is better for small samples and more robust to outliers, than e.g. Pearson's method. Usually, the correlation itself does not disturb the model results and efficiency, although since my goal is to examine the relation between cc_total and GDP_growth, I found it helpful to remove strongly correlated democracy index values. Regarding the correlations presented on the Pic 2. it is worth pointing out that cc_total and GDP_growth have negative rho. This already alarms that the result of the estimation might not follow the expectations. To understand it better I decided to check for the simple correlation results among different groups of countries. The results along with the p-values, given the three different alternative hypotheses, are presented in Table A2. in the Appendix. The creation of groups were dictated by various characteristics: their wealth (where the rich countries are the ones with GDP per capita greater than the median for the whole sample), the continent, their legal origin (historical origin of a country's laws before 1991 - French, Socialist, British, German or Scandinavian) and the fact whether or not they were ever colonized.

Pic 2. Correlation heatmap.



Source: Own study, based on Gutmann et al. (2024), Coppedge et al. (2024) and World Bank (2025)

Despite the correlation being significantly different than zero only for four of the country groups, more concerning is the negative sign of rho in all of those four cases. That would mean that governments should not obey the constitutional laws if they want their country to be prosperous. Table A3. in the Appendix shows the correlations with cc_prop, which according to the theory should have stronger positive effect on economic development, although the correlation results showed only limited improvement. Given these unexpected findings, I proceeded to estimate a regression model. But before doing so, I removed five outliers based on high hat values and high standardized residuals, resulting in a final sample of 157 countries.

In this refined OLS estimation, three variables (and a constant) proved to be statistically significant on their own. When I conducted the F-test for the jointly significance of the remaining variables, p-value indicated no grounds to reject the null hypothesis of all coefficients being equal to 0. Therefore, from an econometric perspective, removing additional variables from the model was justified. I eliminated insignificant variables one-by-one, starting with those that had the highest p-values in the t-test (in the meantime repeating the F-test to make sure it is still justified). I ended up removing only two variables in this way – fertility and inflation, because the next one to be eliminated was our measure of constitutional compliance, which I had interest in leaving in the model. The results are presented in Table 3., where model 1 is our point of departure, and model 2 is its final form.

The diagnostic test results for all of the OLS models are provided in the Table A4. in the Appendix. According to these results, all models exhibit heteroscedasticity, as indicated by the low p-values from the Breusch-Pagan test. This has already been accounted for in Table 3. through the application of robust variance-covariance matrices.

I will focus on the interpretation of Model 2., presented in column (2) of Table 3, as it is more efficient due to the deleted insignificant variables. Surprisingly, constitutional compliance appears to be statistically insignificant, along with terms of trade growth, trade and the interaction of both of them. The coefficients estimated for GDP per capita and government expenditure are both statistically significant and negative. On the other hand, the parameter for investment is highly significant and positive. The constant lacks a meaningful interpretation in all of the models presented in this study, as it would be the GDP growth for a country with 0 initial GDP per capita, 0 babies born per average women etc., however, it as well is highly significant and positive.

Table 3. Estimation results: cross-section model; dependent variable: GDP growth.

	Dependent variable:			
	GDP_growth			
	(1)	(2)	(2lv)	(2prop)
Variables related to constitutional compliance				
log(cc_total + 2)	-0.138	-0.144		
	(0.318)	(0.305)		
log(cc_total_lv + 2)			-0.101	
			(0.458)	
log(cc_prop + 2)				-0.017
				(0.390)
Other control variables				
fertility	0.052			
	(0.215)			
log(GDPpc2015)	-0.537***	-0.574***	-0.590***	-0.585***
	(0.167)	(0.117)	(0.124)	(0.123)
inf_def	0.001			
	(0.053)			
investment	0.099***	0.099***	0.099***	0.100***
	(0.033)	(0.031)	(0.031)	(0.032)
gov_exp_reduced	-0.115**	-0.114***	-0.116***	-0.118***
	(0.046)	(0.043)	(0.044)	(0.044)
tot_growth	0.214	0.226	0.217	0.246
	(0.454)	(0.441)	(0.449)	(0.433)
trade	0.008	0.008	0.008	0.008
	(0.005)	(0.005)	(0.005)	(0.005)
tot_growth:trade	-0.002	-0.002	-0.002	-0.003
	(0.006)	(0.006)	(0.006)	(0.006)
Constant	6.122***	6.597***	6.705***	6.602***
	(2.183)	(1.471)	(1.511)	(1.484)
Observations	122	122	121	122
R^2	0.470	0.469	0.469	0.468
Adjusted R^2	0.427	0.437	0.436	0.435
Residual Std. Error	1.539 (df = 112)	1.526 (df = 114)	1.532 (df = 113)	1.528 (df = 114)

	Dependent variable:			
	GDP_growth			
	(1)	(2)	(2lv)	(2prop)
F Statistic	11.020^{***} (df = 9; 112)	14.393^{***} (df = 7; 114)	14.272^{***} (df = 7; 113)	14.320^{***} (df = 7; 114)
Note:				*p<0.1; **p<0.05; ***p<0.01

Source: Own study.

The results, for the variables that turned out to be significant, are with accordance with the economic growth theory. Namely, conditional convergence, the negative impact of excessive government spending (unproductive consumption) and a clear positive impact of investment on GDP growth are all confirmed.

The lack of significance in constitutional compliance suggests it may have no impact on GDP growth, which contradicts my hypothesis. Notwithstanding, to confirm that result, I decided to reestimate the Model 2 for two other cases: changing the constitutional compliance variable to its equivalent calculated according to Law and Versteeg (2013) method, as well as replacing it with the constitutional compliance only in the area of property rights protection, as this should have the most pronounced impact on economic growth according to the literature. The results for these two models are presented in columns (2lv) and (2prop) of Table 3. The significance and the signs of the coefficients of the significant variables are the same as in Model 2., with constitutional compliance measures being irrelevant for GDP growth rate.

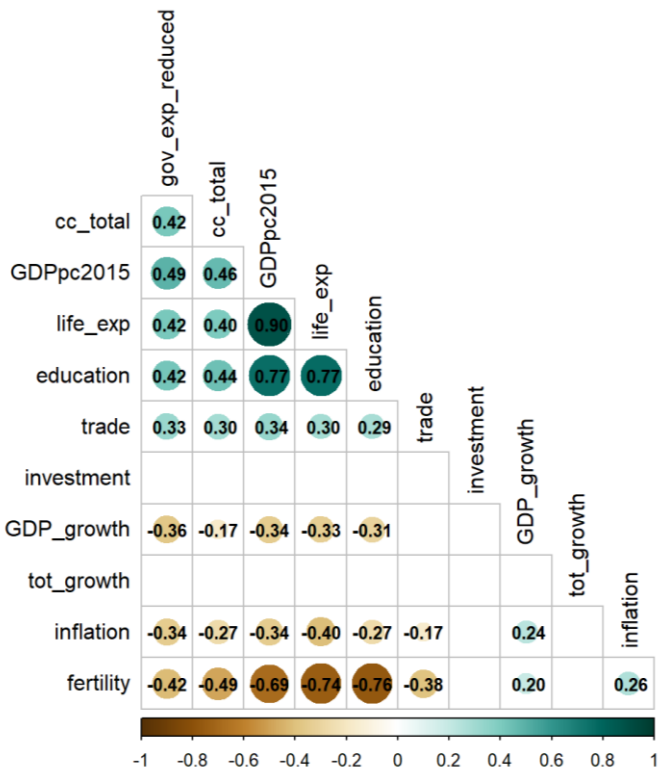
These results, which could have been anticipated given the correlation analysis yet remain surprising, sparked my curiosity and led me to estimate the same model using institutional measures instead (Table A5. in the Appendix) to see what results they would yield. The idea came from the fact that institutions are often seen as important for economic performance and are widely studied, as shown in the literature review. Thus, I wanted to see if the indicators for institutional quality, to which constitutional compliance is closely related, would be important factors in explaining GDP growth in my data. In Models i1-i6, built on the Model 2, the variable cc_total is successively replaced by the following variables: rule of law estimate, control of corruption estimate, government effectiveness estimate, political stability estimate, regulatory quality estimate as well as voice and accountability estimate. This estimation, however, is not as surprising, as five out of six of these institutional measures are statistically significant, having a positive impact on GDP growth. This is in line with the considerations made in the literature review section.

Therefore, the question is, why are the results for the constitutional compliance completely against my hypothesis? The most probable issue is the choice of the model. The OLS estimation might not fully capture the relation between the variables of interest as it fails to introduce the time dimension. That is why, I decided to conduct the panel regression analysis as well.

3.2.2. Panel estimation

I started with enlarging the time dimension of my dataset in order to improve the efficiency of the panel estimation. Pic 3. shows the correlation heatmap for the panel data. As we can see, life expectancy, fertility, education and GDP per capita are again strongly correlated. Moreover, just as before, education variable has a lot of missing values (see Table A6. in the Appendix), while life expectancy has lower than before, but still quite high VIF (around 5.5). For these reason, I decided to remove education and life expectancy from the considerations to keep more observations in the model and to avoid possible multicollinearity.

Pic 3. Correlation heatmap for panel data.



Source: Own study, based on Gutmann et al. (2024) and World Bank (2025)

I used numerous estimation methods, including pooled OLS, between effects, first differences, fixed effects and random effects estimation, to determine the most suitable approach for my data. To guide this selection, I conducted the following specification tests:

- Lagrange Multiplier Test (H0: the pooled OLS is enough as there are no random effects)
- F Test for Individual and/or Time Effects (H0: the pooled OLS is enough as there are no fixed effects)
- Hausman Test (H0: there is no correlation between unobserved effects and regressors, therefore random effects estimator is preferred to the fixed effects one).

The results of these tests are presented in Table A7. in the Appendix. The extremely low p-values suggest that I should stick to the fixed effects estimator in my analysis.

Table 4. presents the estimation results for panel data regressions. While estimating the FE regression, any variables that remain constant for a given country over time do not receive individual coefficient estimates, as they are eliminated during the within-country differencing process. Consequently, for the GDP per capita I use the 2-year lagged values. Most of the variables turn out to be statistically significant and when they do – the sign of the relation between them and GDP growth is in accordance with the macroeconomic theory. However, the issue lies in the efficiency of this model. As presented in the Table A8. in the Appendix, it failed to pass all of the three diagnostic tests:

- Wooldridge Test for autocorrelation
- Breusch-Pagan test for homoscedasticity
- Pesaran CD test for cross-sectional dependence.

This motivated me to use the Driscoll and Kraay (1998) robust standard errors. This method relies on large T asymptotics, which should be appropriate given my 30-year sample period. Table 4. presents the result without using any robust matrices and the result using the Driscoll and Kraay (1998) robust matrix. I initially considered excluding the insignificant variables, but the F-test for their joint significance yielded a very low p-value (with the null hypothesis being that these variables' estimated coefficients are equal to zero). This indicates that from the econometric point of view there is no justification of excluding them from the model, since they proved to be jointly significant.

The results are presented in Table 4; column (FE) reports the fixed effects estimation, while column (DK) presents the robust estimates based on Driscoll and Kraay (1998). The results indicate that the estimated effects are not only statistically significant, but also align with theoretical expectations regarding the direction of the relationships. Specifically, in the robust

model (DK), the estimate of the measure of compliance with constitution is positive and highly significant, which is in line with the predictions based on the literature review. Explicitly, a 1% increase in the cc_total index plus 2 (so the stronger compliance with constitution) is associated with an increase of about 0.01854 p.p. in GDP growth. This suggest that constitutional compliance significantly and positively affects GDP growth.

Moving to the other explanatory variables, the coefficients for investment and trade are positive and significant, while the ones for GDP per capita and government expenditure (reduced by the expenditure on education and defense) are negative and significant, holding other factors constant and controlling for unobserved heterogeneity through fixed constant effects. These results are in line with theoretical considerations. Estimates for openness ratio and inflation are respectively positive and negative, while both being significant at the 10% level. Only parameters for fertility and terms of trade growth are completely not significant.

I also conducted similar regressions, substituting cc_total with the indicators for specific rights category (see Table 5). Unexpectedly, the group of civil rights is the one that has the highest estimated coefficient, surpassing the property rights category. Only the compliance with political rights seem to not have any impact on the rate of GDP growth.

Table 4. Estimation results for panel data regressions; dependent variable: GDP growth.

	Dependent variable:	
	GDP_growth	
	(FE)	(DK)
Variables related to constitutional compliance		
log(cc_total + 2)	1.854***	1.854***
	(0.420)	(0.379)
Other control variables		
fertility	-0.918***	-0.918
	(0.267)	(0.669)
log(lag(GDPpc2015, 2))	-7.396***	-7.396***
	(0.389)	(1.379)
gov_exp_reduced	-0.262***	-0.262***
	(0.040)	(0.082)
inflation	-0.026***	-0.026*
	(0.008)	(0.016)
investment	0.180***	0.180***
	(0.016)	(0.029)
tot_growth	0.003	0.003
	(0.024)	(0.021)
trade	0.053***	0.053***
	(0.005)	(0.010)
tot_growth:trade	0.001**	0.001*
	(0.0003)	(0.0003)
Observations	2,233	2,233
R^2	0.257	0.257
Adjusted R^2	0.206	0.206
F Statistic (df = 9; 2087)	80.320***	80.320***
Note:		*p<0.1; **p<0.05; ***p<0.01

Source: Own study.

Table 5. Estimation results for panel data regressions, grouped by different rights categories.

	Dependent variable:			
	GDP_growth			
	(basic_DK)	(civil_DK)	(polit_DK)	(prop_DK)
Variables related to constitutional compliance				
log(cc_basic + 2)	0.524***			
	(0.124)			
log(cc_civil + 2)		1.912***		
		(0.486)		
log(cc_polit + 2)			0.523	
			(0.437)	
log(cc_prop + 2)				0.993***
				(0.365)
Other control variables				
fertility	-0.872	-1.035	-1.020	-0.918
	(0.680)	(0.689)	(0.679)	(0.692)
log(lag(GDPpc2015, 2))	-7.385***	-7.228***	-7.481***	-7.492***
	(1.397)	(1.375)	(1.393)	(1.393)
gov_exp_reduced	-0.278***	-0.260***	-0.272***	-0.272***
	(0.082)	(0.085)	(0.081)	(0.083)
inflation	-0.027*	-0.027	-0.029*	-0.028*
	(0.015)	(0.016)	(0.017)	(0.016)
investment	0.180***	0.181***	0.181***	0.179***
	(0.029)	(0.028)	(0.029)	(0.029)
tot_growth	0.0004	0.007	-0.001	0.004
	(0.022)	(0.020)	(0.022)	(0.021)
trade	0.054***	0.052***	0.054***	0.054***
	(0.010)	(0.011)	(0.010)	(0.010)
tot_growth:trade	0.001*	0.001*	0.001*	0.001*
	(0.0004)	(0.0003)	(0.0004)	(0.0003)
Observations	2,233	2,233	2,210	2,233
R^2	0.254	0.262	0.253	0.252
Adjusted R^2	0.203	0.211	0.201	0.200
F Statistic	79.138*** (df = 9; 2087)	82.479*** (df = 9; 2087)	77.709*** (df = 9; 2067)	78.227*** (df = 9; 2087)
Note:				*p<0.1; **p<0.05; ***p<0.01

Source: Own study.

3.2.3. Conclusions

To summarize the empirical section of the study, it is evident that more sophisticated econometric techniques, such as panel data regressions, might be more suitable for accurately estimating the impact of constitutional compliance on economic growth. The OLS estimation did not reveal any significant effect of the constitutional compliance index, whether the overall measure or the one related to private property and the rule of law. This outcome may be attributed to OLS's inability to account for time-invariant country-specific effects, which likely influence these constitutional compliance indices. Though this remains a plausible explanation rather than a definitive conclusion. In contrast, the results from the panel data regressions align with expectations, showing that constitutional compliance has a significantly positive effect on GDP growth, consistent with my hypothesis. Throughout the modeling, robust covariance matrices were employed to address issues such as cross-sectional dependence, autocorrelation, and heteroscedasticity. The Driscoll and Kraay (1998) method was used as the most effective for obtaining reliable robust standard errors in case of my panel data modelling.

SUMMARY

This thesis offers an in-depth analysis of constitutional compliance, with a particular focus on its effects on economic growth. The literature review not only examines the gap between laws in theory (de jure) and their implementation in practice (de facto), but also explains why, based on deeper frameworks within institutional and constitutional economics, this gap is expected to have a critical impact on economic outcomes.

Research suggests that the extent to which the government respects the constitution plays a crucial role in a country's economic development. Compliance with the constitutional law might contribute to a broader sense of security among citizens—not only in terms of civil liberties but also economic stability.

Empirical part of this study indicates that adherence to constitutional rights significantly supports economic growth, while disregard for these rights can hinder development, leading to slower GDP growth. Importantly, the simple econometric models as OLS fail to capture this relationship, which signals the need to turn to the more advanced econometric methods, in this case the fixed effects (or within) estimation, to correctly interpret the relationship between economic growth and constitutional compliance. In my research I used Driscoll and Kraay (1998) robust standard errors to cope with the existing model weaknesses, therefore another recommendation for the researchers interested in investigating similar dependencies is to use extended time periods, allowing robust matrices to be used efficiently.

The panel regression results suggest that we cannot reject the hypothesis that constitutional compliance positively influences GDP growth, even though the OLS model does not yield statistically significant results, likely due to its inability to account for time-invariant country-specific effects, which the panel approach addresses. Against prior assumptions, civil rights such as free movement, religious freedom, free speech and free media reveal a more substantial impact on GDP growth than the property rights and rule of law. It is surprising, since the latter are commonly perceived as the ones influencing the investors and their sense of safety, which should exert more significant pressure on economic growth rates. This is certainly a very interesting topic to expand and explore in further research.

In light of this findings, governments and politicians should focus on strengthening institutions responsible for compliance with constitutions, such as independent constitutional courts, to ensure greater consistency of government actions with the rule of law. Increasing transparency of government actions would also encourage building trust of the investors and

citizens. Of great importance is respect for the fundamental civil rights which affect the stability and predictability of political environment.

The presented study is one of the few works on the topic of constitutional compliance impact on economic outcomes, which makes it particularly valuable in the context of current knowledge. It incorporates the latest data on constitutional compliance provided by Gutmann et al. (2024) and employs cross-sectional as well as the panel data analysis, offering an insightful perspective on the issue and making a meaningful contribution to the understanding of the topic of de jure-de facto gap consequences.

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APPENDIX. Supplementary Analyses and Data Information.

Table A1. Descriptive statistics for the period 2010-2019.

Variable	N	Min	Median	Mean	SD	Max
GDP_growth	156	-6.36	3.47	3.50	2.33	9.78
fertility	157	1.15	2.33	2.83	1.39	6.97
GDPpc2015	154	289.86	4601.97	11681.08	16314.35	81315.42
gov_exp	145	5.51	15.54	16.03	5.56	37.83
gov_exp_edu	145	0.46	4.36	4.41	1.61	10.24
gov_exp_milit	142	0.05	1.40	1.90	1.52	9.18
inf_def	156	-0.25	3.45	5.67	6.75	44.90
investment	144	9.69	23.54	24.63	7.45	58.82
life_exp	157	46.02	72.01	69.73	8.90	82.84
education	113	0.74	30.98	36.46	27.67	101.69
tot_growth	156	-4.18	0.48	0.54	1.29	4.02
trade	145	22.93	73.92	82.28	41.24	293.73
cc_basic	157	-1.99	0.53	0.32	1.07	1.59
cc_civil	157	-1.67	0.76	0.28	1.03	1.40
cc_polit	154	-1.41	0.64	0.41	1.00	1.81
cc_prop	157	-1.41	-0.05	0.21	1.12	2.01
cc_total	157	-1.73	0.58	0.35	1.09	1.97
cc_total_lv	156	-1.27	0.56	0.34	0.91	1.41
v2x_libdem	157	0.01	0.38	0.42	0.27	0.89
gov_exp_reduced	124	-0.26	9.22	9.48	4.19	19.19
ctrl_of_corr	157	-1.69	-0.42	-0.15	0.99	2.26
gov_eff	157	-2.24	-0.24	-0.11	0.97	2.05
pol_stab	157	-2.65	-0.14	-0.20	0.92	1.44
reg_q	157	-2.40	-0.23	-0.09	0.97	1.93
rule_of_law	157	-2.36	-0.35	-0.16	0.97	2.01
voice_and_acc	157	-2.22	-0.18	-0.16	0.99	1.68

Source: Own study, based on Gutmann et al. (2024), Coppedge et al. (2024) and World Bank (2025)

Table A2. Spearman's rho Correlation Test Results: GDP_growth vs cc_total, H0: rho = 0

			P-value		
	N	**rho**	**Alternative Two-Sided**	**Alternative Less**	**Alternative Greater**
complete data	157	-0.2614	0.0010	0.0005	0.9995
rich	79	-0.1722	0.1316	0.0658	0.9342
poor	78	-0.1277	0.2646	0.1323	0.8677
Africa	48	0.0363	0.8063	0.4031	0.5969
America	27	-0.0885	0.6595	0.3298	0.6702
Asia	41	-0.0548	0.7363	0.3682	0.6318
Europe	35	0.0282	0.8724	0.4362	0.5638
Pacific	6	0.0286	1.0000	0.5000	0.5000
FR	72	-0.1181	0.3267	0.1633	0.8367
SO	32	-0.3193	0.0855	0.0427	0.9573
UK	45	-0.2280	0.1363	0.0681	0.9319
GE	6	0.1000	0.9500	0.4750	0.5250
SC	6	-0.2000	0.7833	0.3917	0.6083
ever_colonized	130	-0.1998	0.0232	0.0116	0.9884
never_colonized	27	-0.5706	0.0019	0.0009	0.9991

Source: Own study, based on Gutmann et al. (2024), Coppedge et al. (2024), Conte et al. (2022) and World Bank (2025)

Table A3. Spearman's rho Correlation Test Results: GDP_growth vs cc_prop, H0: rho = 0

			P-value		
	N	**rho**	**Alternative Two-Sided**	**Alternative Less**	**Alternative Greater**
complete data	157	-0.2529	0.0014	0.0007	0.9993
rich	79	-0.2149	0.0589	0.0294	0.9706
poor	78	-0.0472	0.6815	0.3407	0.6593
Africa	48	0.0202	0.8918	0.4459	0.5541
America	27	-0.2172	0.2765	0.1382	0.8618
Asia	41	-0.0925	0.5702	0.2851	0.7149
Europe	35	-0.1153	0.5097	0.2548	0.7452
Pacific	6	-0.6571	0.1750	0.0875	0.9125
FR	72	-0.1467	0.2220	0.1110	0.8890
SO	32	-0.2780	0.1369	0.0685	0.9315
UK	45	-0.1941	0.2067	0.1033	0.8967
GE	6	0.5270	0.3615	0.1807	0.8193
SC	6	-0.7000	0.2333	0.1167	0.8833
ever_colonized	130	-0.1699	0.0542	0.0271	0.9729
never_colonized	27	-0.4598	0.0158	0.0079	0.9921

Source: Own study, based on Gutmann et al. (2024), Coppedge et al. (2024), Conte et al. (2022) and World Bank (2025)

Table A4. Regression Diagnostic Tests.

	P-value			
Model	**RESET**	**Shapiro-Wilk**	**Breusch-Pagan**	**Durbin-Watson**
1	0.9070	0.1664	0.0015	0.4457
2	0.9874	0.1748	0.0125	0.4895
2lv	0.9494	0.2403	0.0087	0.4902
2prop	0.8615	0.2635	0.0027	0.5000
i1	0.9665	0.2344	0.0003	0.3961
i2	0.8047	0.1771	0.0163	0.3416
i3	0.6075	0.3365	0.0152	0.3889
i4	0.9017	0.3857	0.0046	0.4770
i5	0.4330	0.4453	0.0001	0.4283
i6	0.6554	0.3386	0.0012	0.5017

Source: Own study.

Table A5. Estimation results. Institutional variables included.

	Dependent variable:					
	GDP_growth					
	(i1)	(i2)	(i3)	(i4)	(i5)	(i6)
log(GDPpc2015)	-0.980***	-0.986***	-1.178***	-0.829***	-0.875***	-0.623***
	(0.202)	(0.179)	(0.229)	(0.144)	(0.206)	(0.151)
investment	0.093***	0.093***	0.089***	0.097***	0.102***	0.102***
	(0.033)	(0.033)	(0.032)	(0.031)	(0.032)	(0.032)
gov_exp_reduced	-0.144***	-0.161***	-0.145***	-0.148***	-0.128***	-0.124***
	(0.045)	(0.043)	(0.043)	(0.040)	(0.047)	(0.046)
rule_of_law	0.792**					
	(0.309)					
ctrl_of_corr		0.835***				
		(0.236)				
gov_eff			1.140***			
			(0.338)			
pol_stab				0.765**		
				(0.302)		
reg_q					0.627*	
					(0.376)	
voice_and_acc						0.111
						(0.264)
tot_growth	0.195	0.225	0.180	0.301	0.227	0.240
	(0.401)	(0.391)	(0.418)	(0.389)	(0.413)	(0.427)
trade	0.007	0.008*	0.007	0.005	0.006	0.008
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
tot_growth:trade	-0.001	-0.001	-0.001	-0.003	-0.002	-0.002
	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)
Constant	10.432***	10.606***	12.215***	9.347***	9.183***	6.932***
	(2.229)	(1.988)	(2.383)	(1.715)	(2.164)	(1.745)
Observations	122	122	122	122	122	122
R^2	0.505	0.517	0.525	0.509	0.493	0.469
Adjusted R^2	0.475	0.488	0.496	0.479	0.462	0.437
Residual Std. Error (df = 114)	1.473	1.455	1.444	1.469	1.492	1.526
F Statistic (df = 7; 114)	16.647***	17.464***	17.981***	16.861***	15.833***	14.397***

Note: *p<0.1; **p<0.05; ***p<0.01

Source: Own study.

Table A6. Descriptive statistics for the period 1990-2020.

Variable	N	Min	Median	Mean	SD	Max
cc_basic	4786	-1.99	0.50	0.26	1.12	1.59
cc_civil	4786	-1.67	0.90	0.30	1.03	1.40
cc_polit	4668	-1.41	0.45	0.37	1.04	1.81
cc_prop	4771	-1.41	-0.05	0.19	1.13	2.01
cc_total	4786	-1.86	0.45	0.32	1.08	1.97
cc_total_lv	4738	-1.27	0.52	0.32	0.94	1.41
education	3198	0.00	28.00	34.02	27.12	143.96
fertility	4786	0.84	2.60	3.14	1.67	8.46
GDP_growth	4702	-51.03	3.75	3.46	6.41	149.97
gov_exp_reduced	2638	-30.75	9.63	9.73	4.65	39.70
inflation	4700	-31.57	4.78	31.32	316.98	15444.42
investment	4212	-3.95	22.71	23.65	8.33	76.78
tot_growth	3910	-62.29	0.01	0.92	11.55	349.44
trade	4266	9.96	70.21	81.23	48.91	437.33
GDPpc2015	4682	188.66	3953.10	11608.53	17055.86	112417.88
ever_colonized	4786	0.00	1.00	0.83	0.37	1.00
life_exp	4786	12.16	70.33	68.39	9.62	84.56

Source: Own study, based on Gutmann et al. (2024) and World Bank (2025)

Table A7. P-Values for Panel Regression Specification Tests

Test	P-Value
Lagrange Multiplier Test	0.0000
F Test for Individual and/or Time Effects	0.0000
Hausman Test	0.0000

Source: Own study.

Table A8. P-Values for Panel Regression Diagnostic Tests

Test	P-Value
Wooldridge Test	0.0000
Breusch-Pagan	0.0000
Pesaran CD	0.0000

Source: Own study.